

Feature Selection

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Selection addresses

- Cost
 - Speed
 - Simplicity
 - Robustness
-
- You can keep only 10 features. Which survive?

Automatic feature generation

Filter methods

- Statistical analysis
- No modeling
- Fast

Wrapper methods

- RFECV
- Forwards selection
- Stepwise selection

~~Wrapper methods~~



Embedded methods

- RFECV
- Forwards selection
- Stepwise selection

Selection methods

Method	Speed	Accuracy	Complexity	Best Use Case
Filter	Very Fast	Moderate	Low	Quick data cleaning / Big Data
Wrapper	Slow	High	High	Small-to-medium data / Peak performance
Embedded	Moderate	High	Moderate	Regularized models (Lasso, Trees)

Classification metrics

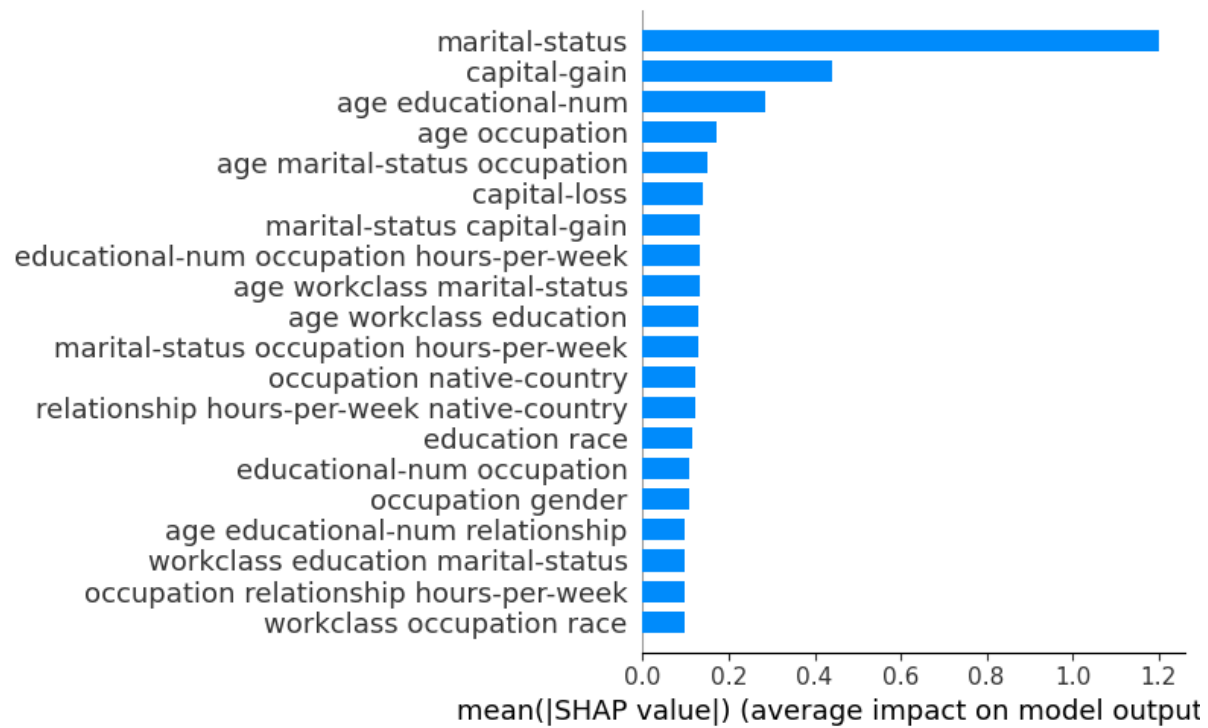
Metric	What it measures	Pros	Cons	When to use
Accuracy	Overall correctness	Simple	Misleading with imbalance	Balanced datasets
Precision	Correctness of positive predictions	Controls false positives	Ignores false negatives	When FP is costly
Recall	Coverage of positives	Finds all positives	Ignores false positives	When FN is costly
F1 Score	Balance of precision & recall	Single number	Hides tradeoff	General summary
Balanced Accuracy	Avg recall per class (0.5=random guess)	Handles imbalance well	Less intuitive	Imbalanced data
ROC AUC	Ranking ability (0.5=random guess)	Threshold-independent	Hard to interpret	Model comparison

Shap

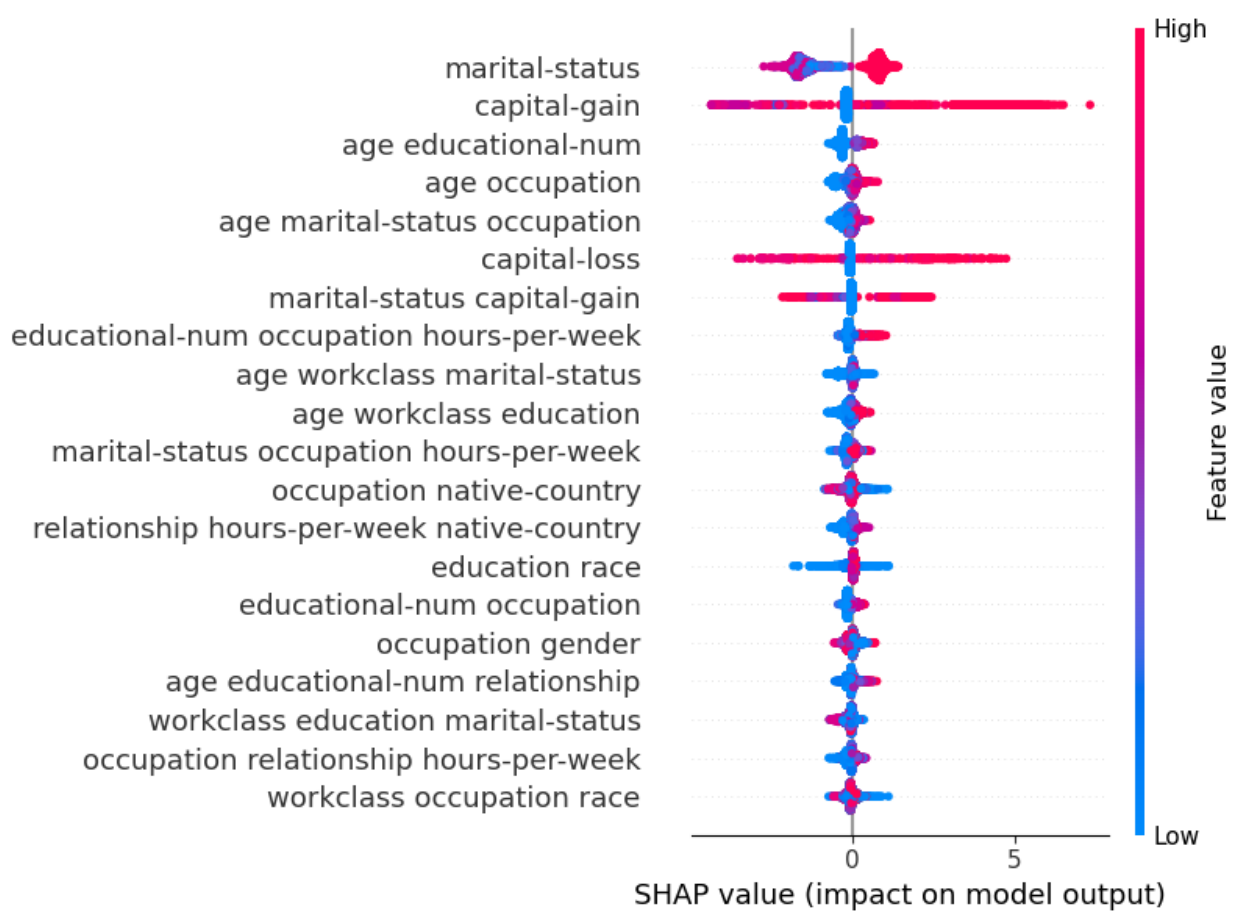
$$\phi_0 = \mathbb{E}[f(X)]$$

$$\text{prediction} = \text{baseline} + \sum \text{feature contributions}$$

SHAP barplot

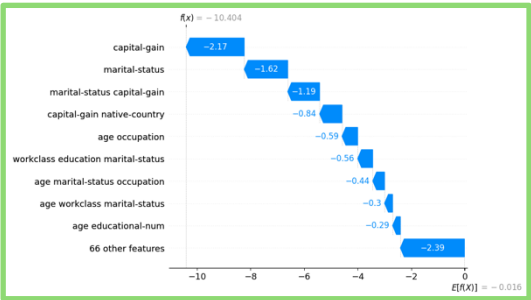
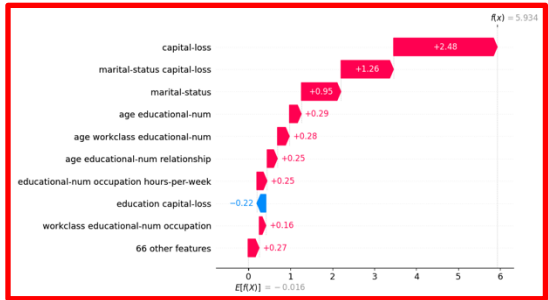
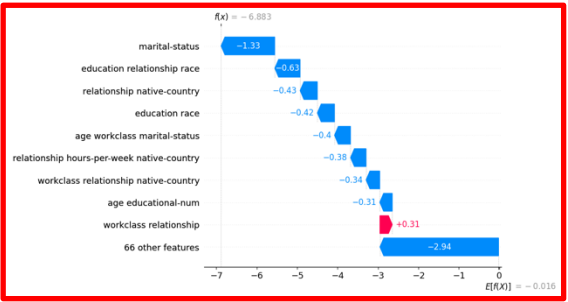
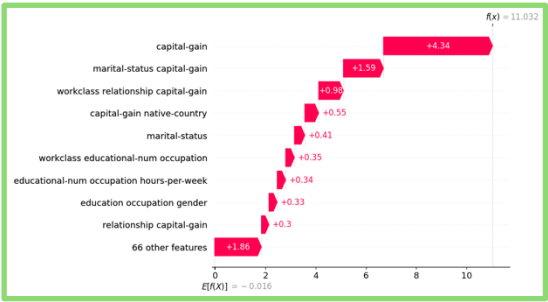


SHAP bee swarm



SHAP waterfall plots

		Predicted	
		1	0
Actual	1	6182 (TP)	1249 (FN)
	0	381 (FP)	1957 (TN)



Select features using multiple tools

	Feature Importance (Tree)	Permutation Importance	SHAP
What it calculates	Split usage / gain / impurity reduction.	Change in model performance when feature is shuffled.	Average magnitude of feature contributions.
What it means	What the model uses during training.	What the model depends on for prediction.	How much a feature changes predictions.
Strengths	Fast. Built-in. Easy to compute.	Directly tied to performance. Model-agnostic. Simple concept	Interpretable. Shows direction (+/-). Captures interactions.
Weaknesses	Biased (e.g., high cardinality). Not tied to performance. Can inflate importance.	Underestimates correlated features. Requires repeated predictions. No direction.	Not tied to performance. Splits importance across correlated features. More computationally expensive.
Best use	Initial ranking	Feature selection	Interpretation / explanation

Stacking



Blending/stacking

- Averaging of predictions from multiple models (or weighted average)
- Meta-learner created using outputs from multiple models

PyCaret

